

15.20 This exercise uses data from the STAR experiment introduced to illustrate fixed and random effects for grouped data. In the STAR experiment, children were randomly assigned within schools into three types of classes: small classes with 13–17 students, regular-sized classes with 22–25 students, and regular-sized classes with a full-time teacher aide to assist the teacher. Student scores on achievement tests were recorded as well as some information about the students, teachers, and schools. Data for the kindergarten classes are contained in the data file *star*.

- a. Estimate a regression equation (with no fixed or random effects) where *READSCORE* is related to *SMALL*, *AIDE*, *TCHEXPER*, *BOY*, *WHITE_ASIAN*, and *FREELUNCH*. Discuss the results. Do students perform better in reading when they are in small classes? Does a teacher's aide improve scores? Do the students of more experienced teachers score higher on reading tests? Does the student's sex or race make a difference?
- b. Reestimate the model in part (a) with school fixed effects. Compare the results with those in part (a). Have any of your conclusions changed? [Hint: specify *SCHID* as the cross-section identifier and *ID* as the "time" identifier.]
- c. Test for the significance of the school fixed effects. Under what conditions would we expect the inclusion of significant fixed effects to have little influence on the coefficient estimates of the remaining variables?
- d. Reestimate the model in part (a) with school random effects. Compare the results with those from parts (a) and (b). Are there any variables in the equation that might be correlated with the school effects? Use the LM test for the presence of random effects.
- e. Using the *t*-test statistic in equation (15.36) and a 5% significance level, test whether there are any significant differences between the fixed effects and random effects estimates of the coefficients on *SMALL*, *AIDE*, *TCHEXPER*, *WHITE_ASIAN*, and *FREELUNCH*. What are the implications of the test outcomes? What happens if we apply the test to the fixed and random effects estimates of the coefficient on *BOY*?
- f. Create school-averages of the variables and carry out the Mundlak test for correlation between them and the unobserved heterogeneity.

(a)

1. 學生在小班 (*small*) 表現更好嗎？

small 的估計係數為 5.82282，其 *t* 值為 5.886，*p*-value 為 4.19e-09（遠小於 0.05）。

是的，表現顯著更好。在控制其他條件不變下，被隨機分配到小班（13–17 人）的學生，其閱讀測驗分數平均會比待在一般大班的學生高出約 5.82 分，這在統計上與實質效果上都非常顯著。

2. 助教 (*aide*) 能提高分數嗎？

aide 的估計係數為 0.81784，*p*-value 為 0.391（遠大於 0.05），無顯著標記。

不能，效果不顯著。雖然係數符號為正，但因為 *p* 值相當高，我們無法拒絕「助教效果為零」的虛無假設。這意味著在一般大班配置一名全職教師助教，對於提升學生的閱讀成績並沒有顯著的統計幫助。

3. 資深教師 (*tchexper*) 的學生分數較高嗎？

tchexper 的估計係數為 0.49247，*p*-value 為 1.61e-12（遠小於 0.05）。

是的，分數較高。教師的教學經驗每增加一年，學生的閱讀分數預期會顯著提升約 0.49 分。這說明資深教師累積的教學技巧對兒童早期的閱讀發展有顯著的正向貢獻。

4. 學生的性別 (*boy*) 有帶來差異嗎？

boy 的估計係數為 -6.15642，*p*-value 為 1.23e-14（遠小於 0.05），帶有 *** 極顯著標記。

有顯著差異，男生表現較弱。在同等條件下（不論班級類型或家庭背景），男生的閱讀測驗平均分數比女生低了約 6.16 分。

5. 學生的種族 (white_asian) 有帶來差異嗎？

white_asian 的估計係數為 3.90581，p-value 為 4.26e-05（遠小於 0.05）。有顯著差異，白人與亞裔學生分數較高。在其他條件控制不變下，白人或亞裔學生的閱讀分數比其他族裔（如非裔）學生平均顯著高出約 3.91 分。

```
=== (a) OLS 迴歸模型估計結果 ===
> model_a <- lm(readscore ~ small + aide + tchexper + boy + white_asian + freelunch, data = star)
> print(summary(model_a))

Call:
lm(formula = readscore ~ small + aide + tchexper + boy + white_asian +
    freelunch, data = star)

Residuals:
    Min       1Q   Median       3Q      Max
-107.220  -20.214   -3.935   14.339  185.956

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  437.76425    1.34622  325.180 < 2e-16 ***
small         5.82282    0.98933   5.886 4.19e-09 ***
aide          0.81784    0.95299   0.858  0.391
tchexper      0.49247    0.06956   7.080 1.61e-12 ***
boy          -6.15642    0.79613  -7.733 1.23e-14 ***
white_asian   3.90581    0.95361   4.096 4.26e-05 ***
freelunch    -14.77134    0.89025 -16.592 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 30.19 on 5759 degrees of freedom
(因為不存在，20 個觀察量被刪除了)
Multiple R-squared:  0.09685,    Adjusted R-squared:  0.09591
F-statistic: 102.9 on 6 and 5759 DF,  p-value: < 2.2e-16
```

(b)

小班制效果 (small) 的結論更為穩健：在 (a) 中，小班制能提升約 5.82 分；當我們在 (b) 控制了學校固定效果（排除了學校硬體、學區資源等未觀察異質性）後，小班制的效果反而提高到了 6.49 分。這證明小班制確實能有效改善學習成效。

遺漏變數偏誤 (Omitted Variable Bias) 的證據：老師經驗 (tchexper) 的係數從 0.49 降到 0.29。這說明在 OLS 模型中，老師經驗的效應被高估了。因為教學經驗豐富的老師往往聚集在資源較好的學校，OLS 會把「學校環境好」的功勞誤記在「老師資深」上。

而種族的係數大幅翻倍上升且極度顯著。說明在同一個學校內部，白人與亞裔學生的優勢比跨校比較時更為明顯。

```

>
> print(summary(model_b))
Oneway (individual) effect within Model

Call:
plm(formula = readscore ~ small + aide + tchexper + boy + white_asian +
    freelunch, data = p_star, effect = "individual", model = "within")

Unbalanced Panel: n = 79, T = 34-137, N = 5766

Residuals:
    Min.     1st Qu.     Median     3rd Qu.     Max.
-102.6381  -16.7834   -2.8473   12.7591   198.4169

Coefficients:
              Estimate Std. Error t-value Pr(>|t|)
small         6.490231    0.912962   7.1090 1.313e-12 ***
aide          0.996087    0.881693   1.1297  0.2586
tchexper      0.285567    0.070845   4.0309 5.629e-05 ***
boy          -5.455941    0.727589  -7.4987 7.440e-14 ***
white_asian   8.028019    1.535656   5.2277 1.777e-07 ***
freelunch    -14.593572    0.880006  -16.5835 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares:    4628000
Residual Sum of Squares: 4268900

```

(c)

由於 p-value 遠小於 5% 的顯著水準，拒絕虛無假設（所有學校的固定效果皆為 0）。證據表明，各所學校之間確實存在顯著且不可忽略的未觀察異質性（如校風、學區資源、硬體設備差異等）。因此，使用普通的混合 OLS 模型會忽略這些群組特徵，在計量經濟學理論上，我們必須採用 (b) 小題的學校固定效果模型。

```

> print(joint_test)

F test for individual effects

data:  readscore ~ small + aide + tchexper + boy + white_asian + freelunch
F = 16.698, df1 = 78, df2 = 5681, p-value < 2.2e-16
alternative hypothesis: significant effects

```

(d)

```
>
> print(summary(model_d))
Oneway (individual) effect Random Effect Model
(Swamy-Arora's transformation)

Call:
plm(formula = readscore ~ small + aide + tchexper + boy + white_asian +
    freelunch, data = p_star, effect = "individual", model = "random",
    random.method = "swar")

Unbalanced Panel: n = 79, T = 34-137, N = 5766

Effects:
              var std.dev share
idiosyncratic 751.43   27.41 0.829
individual    155.31   12.46 0.171
theta:
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.6470  0.7225  0.7523  0.7541  0.7831  0.8153

Residuals:
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
-97.4830 -17.2361  -3.2822   0.0372  12.8027  192.3460

Coefficients:
              Estimate Std. Error z-value Pr(>|z|)
(Intercept) 436.126774   2.064782 211.2217 < 2.2e-16 ***
small        6.458722   0.912548   7.0777 1.466e-12 ***
aide         0.992146   0.881159   1.1260  0.2602
tchexper     0.302679   0.070292   4.3060 1.662e-05 ***
boy         -5.512081   0.727639  -7.5753 3.583e-14 ***
white_asian  7.350477   1.431376   5.1353 2.818e-07 ***
freelunch   -14.584332   0.874676 -16.6740 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares:    6158000
Residual Sum of Squares: 4332100
R-Squared:              0.29655
Adj. R-Squared:         0.29582
Chisq: 493.205 on 6 DF, p-value: < 2.22e-16

> print(lm_test)

Lagrange Multiplier Test - (Breusch-Pagan)

data:  readscore ~ small + aide + tchexper + boy + white_asian + freelunch
chisq = 6677.4, df = 1, p-value < 2.2e-16
alternative hypothesis: significant effects

>
>
```

```

=== (a) OLS 迴歸模型估計結果 ===
> model_a <- lm(readscore ~ small +
> print(summary(model_a))

Call:
lm(formula = readscore ~ small +
    freelunch, data = star)

Residuals:
    Min       1Q   Median       3Q      Max
-107.220  -20.214   -3.935   14.311  107.220

Coefficients:
            Estimate Std. Error
(Intercept) 437.76425    1.34622
small        5.82282    0.98933
aide         0.81784    0.95299
tchexper     0.49247    0.06956
boy         -6.15642    0.79613
white_asian  3.90581    0.95361
freelunch   -14.77134    0.89025

---
> print(summary(model_b))
Oneway (individual) effect within l

Call:
plm(formula = readscore ~ small + i
    freelunch, data = p_star, effe

Unbalanced Panel: n = 79, T = 34-1

Residuals:
    Min.    1st Qu.    Median    3rd Qu.    Max.
-102.6381  -16.7834   -2.8473    12.3111   102.6381

Coefficients:
            Estimate Std. Error
small        6.490231    0.912962
aide         0.996087    0.881693
tchexper     0.285567    0.070845
boy         -5.455941    0.727589
white_asian  8.028019    1.535656
freelunch   -14.593572    0.880006

```

Column (b) FE 與 **Column (d) RE** 的數值極度接近，暗示隨機效果模型的係數偏誤（偏離不偏估計量）可能不算太嚴重，但仍有些許差異。

white_asian（種族比例）與 **freelunch**（學校家庭經濟弱勢比例）還有 **tchexper**（明星教師資源）是最容易與學校效果（ u_i ）發生相關性。

檢定統計量與 p 值： LM (Chisq)= 6677.4，df = 1，p-value < $2.2 * 10^{-16}$ 。

由於 p 值顯著小於 5% 顯著水準，拒絕虛無假設（ $H_0: \sigma_u^2 = 0$ ）。

證明學校之間存在未觀察的群組異質性，不應該使用傳統的混合 OLS 模型（Pooled OLS）。

Logic of the **Lagrange Multiplier** test for random effects

If $var(u_i) = 0$, u_i is a constant of value zero, there is no random individual heterogeneity.

H_0 : No random effects v.s. H_1 : Random effects exist

$\Leftrightarrow H_0 : u_i = 0$ v.s. $H_1 : u_i \neq 0$

$\Leftrightarrow H_0 : \sigma_u^2 = 0$ v.s. $H_1 : \sigma_u^2 > 0$

$\Leftrightarrow H_0 : cov(\nu_{it}, \nu_{is}) = 0$ v.s. $H_1 : cov(\nu_{it}, \nu_{is}) > 0$

(e)

```
=== (e) (15.36) 各變數之 Hausman t 檢定 ===
>
> # 1. 提取 FE (model_b) 與 RE (model_d) 的係數向量 (RE 需扣除截距項)
> b_fe <- coef(model_b)
> b_re <- coef(model_d)[-1]
>
> # 2. 提取兩模型的對角線標準誤平方 (變異數)
> se_var_fe <- diag(vcov(model_b))
> se_var_re <- diag(vcov(model_d))[-1]
>
> # 3.
> denominator <- sqrt(se_var_fe - se_var_re)
警告訊息：
於 sqrt(se_var_fe - se_var_re)：產生了 NaNs
> hausman_t <- (b_fe - b_re) / denominator
>
> # 4. 算出 z 分配雙尾檢定 p 值
> hausman_p <- 2 * (1 - pnorm(abs(hausman_t)))
>
> # 5. 整合輸出對照表
> hausman_t_summary <- data.frame(
+   Beta_FE = b_fe,
+   Beta_RE = b_re,
+   t_stat = hausman_t,
+   p_value = hausman_p
+ )
> print(round(hausman_t_summary, 5))
      Beta_FE Beta_RE t_stat p_value
small      6.49023  6.45872  1.14601 0.25179
aide       0.99609  0.99215  0.12844 0.89780
tchexper   0.28557  0.30268 -1.93772 0.05266
boy        -5.45594 -5.51208      NaN      NaN
white_asian 8.02802  7.35048  1.21807 0.22320
freelunch  -14.59357 -14.58433 -0.09555 0.92388
```

當程式跑到 boy 這個變數時，因為 se_var_fe (0.727589^2) 小於 se_var_re (0.727639^2)，相減變成負數。沒辦法對負數開根號，所以這一步會產生 NaN。

(f)

學校效果不具備內生性：加入的學校特質平均項 (mean_ 變數) 集體而言對模型並不具備顯著的額外解釋力，說明學校未觀察到的個體效果 (Unobserved school-specific heterogeneity, c_i) 與自變數之間沒有系統性的關聯 (即不存在內生性問題)。

```
~ R 4.5.2 · C:/Users/user/Desktop/計算 /
> # 2
> mundlak <- lm(readscore ~ small + aide + tchexper + boy + white_asian + freelunch +
+   mean_small + mean_aide + mean_tchexper + mean_freelunch,
+   data = star)
>
> robust_vcov <- vcovCL(mundlak, cluster = ~schid)
>
> # 3.
> cat("\n--- OLS 迴歸估計結果 (穩健標準誤) ---\n")
--- OLS 迴歸估計結果 (穩健標準誤) ---
> print(coeftest(mundlak, vcov = robust_vcov))

t test of coefficients:

      Estimate Std. Error t value Pr(>|t|)
(Intercept)  432.58566   15.07626  28.6932 < 2.2e-16 ***
small         6.57140    1.66298   3.9516 7.858e-05 ***
aide          1.07422    1.40159   0.7664 0.44345
tchexper      0.28967    0.13844   2.0925 0.03644 *
boy          -6.06843    0.74659  -8.1282 5.303e-16 ***
white_asian   3.50134    2.72318   1.2858 0.19858
freelunch    -15.13468    1.05431 -14.3550 < 2.2e-16 ***
mean_small   -21.67352   22.44946  -0.9654 0.33437
mean_aide     8.97460   20.23866   0.4434 0.65747
mean_tchexper 1.05851    0.56388   1.8772 0.06054 .
mean_freelunch 1.47474    5.61557   0.2626 0.79286
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

>
```

```

=== (f) 執行 wald 檢定 (Mundlak 檢定) ===
> # 4 聯合 wald 檢定，檢定 4 個平均值項目的斜率是否聯合為 0
> mundlak_wald <- linearHypothesis(
+   mundlak,
+   c("mean_small = 0",
+     "mean_aide = 0",
+     "mean_tchexper = 0",
+     "mean_freelunch = 0"),
+   vcov = robust_vcov
+ )
>
> print(mundlak_wald)

Linear hypothesis test:
mean_small = 0
mean_aide = 0
mean_tchexper = 0
mean_freelunch = 0

Model 1: restricted model
Model 2: readscore ~ small + aide + tchexper + boy + white_asian + freelunch +
  mean_small + mean_aide + mean_tchexper + mean_freelunch

Note: Coefficient covariance matrix supplied.

   Res.Df Df    F Pr(>F)
1     5695
2     5691  4 1.508 0.1969
> |

```